

Supporting Information for

Evolution of Loess Tablelands in the Central Great Plains: Relief Generation by Loess Accumulation and the Importance of Closed Depressions

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Introduction

This supporting information document provides more detail on methods and additional figures and animations illustrating results discussed in the paper. It includes Python and R code used in modeling and analyses and two tables of data.

Text S1: Reconstruction of bedrock surfaces

Within rectangular bounding boxes representing each of the three study areas, we extracted shapefiles with point locations for registered wells from the statewide dataset obtained from the Nebraska Department of Natural Resources (NeDNR)(<https://www.nebraskamap.gov/datasets/nebraska::groundwater-wells-dnr/about>) and using the well registration number, located a driller's log for each well in the NeDNR database. Similarly, we extracted point locations and associated data from a shapefile and database of Conservation and Survey Division (CSD) geologic testholes (<https://snr.unl.edu/csd/geology/testholes.aspx>); the database includes information transcribed from a geologist's log of lithology and stratigraphy. The elevation of each selected registered well or test hole was extracted from a 10-m DEM based on the point locations in the shapefiles. In two study areas with abundant registered wells (Enders Table, Wauneta Table) we used the CSD geologic testholes only as a check on our interpretations of nearby registered wells. In the third study area with sparse registered wells (Ogallala_W), we combined registered well logs and geologic testholes to interpolate the bedrock surface.

The top of the Ogallala Group or the Broadwater Formation (where present) was identified by depth in the registered well driller's logs. Identification of the Ogallala Grp. was based on carbonate cementation, rhizoliths ("rootlets" in logs), gravel or sand beds, and color, drawing on Mason's prior experience with drilling and stratigraphic interpretation of drill cuttings in this region, and referring as needed to nearby geologic testholes with more consistent lithological descriptions done by a geologist. The Broadwater Formation was identified from the presence of gravel and in particular, red or pink colors characteristic of this unit where derived from granites of the Colorado Front Range. Generally the top of both units was easily identified, and the relatively few drill holes where these boundaries could not be identified were removed from the data set. Elevations corresponding to our identified upper Ogallala or Broadwater boundaries were calculated.

Overlaying the drill hole locations on elevation and hillshade layers, we removed those on mid- to lower table margin slopes, leaving points mostly on bedrock or loess tableland summits. The logic behind this step is that the Ogallala Grp. and/or Broadwater Fm. could have been eroded on table margin slopes during stream incision, and if so

would not represent the bedrock tableland surfaces we were tracing beneath loess tablelands. An interpolated surface of the top of the Ogallala Grp. or Broadwater Fm. was then produced by inverse distance weighted interpolation using the second power of distance and the 20 nearest points. Methods are described in the paper text. Finally, the interpolated thickness was clipped to match the extant summits of bedrock or loess tablelands.

The final shapefiles used in interpolation for each area are archived at Mason (2025) <https://doi.org/10.5281/zenodo.14867711> and included here in Data Set 1.

Text S2. Analysis and plotting of strength measurements

Strength measurements (Table S2, uploaded separately) were plotted and statistical tests were performed on them using R (R Core Team, 2022) with the script *strength_plots.R*, included here as supporting data and archived at Mason (2025), <https://doi.org/10.5281/zenodo.14867711>. This script uses ggplot2 3.4.3 and cowplot 1.1.3 packages. Used to produce Figure 5 and the following statistical test results.

Pearson's product-moment correlation, pocket penetrometer and Torvane
data: TV_strength2\$Strength and TV_strength2\$PP_Strength
t = 2.2354, df = 25, p-value = 0.03456
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
0.03329807 0.68233073
sample estimates:
cor 0.4081474

Kruskal-Wallis rank sum test, pocket penetrometer
data: Strength by Unit
Kruskal-Wallis chi-squared = 8.574, df = 6, p-value = 0.199
data: Strength by Horizon
Kruskal-Wallis chi-squared = 0.64168, df = 1, p-value = 0.423

Kruskal-Wallis rank sum tests, Torvane
data: Strength by Unit
Kruskal-Wallis chi-squared = 5.2049, df = 6, p-value = 0.5178
data: Strength by Horizon
Kruskal-Wallis chi-squared = 0.35875, df = 1, p-value = 0.5492

Text S3. Implementation of landscape evolution modeling

The landscape evolution modeling described in the paper used two Python scripts, *Loess_Tablelands_v1.py* and *Loess_Tablelands_v2.py*, included here as separate files in Data Set 2 and also archived at Mason (2025), <https://doi.org/10.5281/zenodo.14867711>. The scripts were run most recently with Python 3.9.15. Both scripts include the following from the Landlab 2.7 Python library (Hobley et al., 2017; Barnhart et al., 2020):

```
landlab:
    RasterModelGrid
landlab.components:
    LinearDiffuser
    FlowAccumulator
    ErosionDeposition
landlab.io
    write.esri.ascii
    read.esri.ascii
landlab.plot
    imshow.grid
```

The scripts also use numpy and os. *Loess_Tablelands_v1.py* reads one of two DEMs of bedrock tableland surfaces (og_surf4.txt for the Bedrock-Depressions scenario, og_surf6.txt for the Bedrock-Drain scenario, Data Set 2) and uses it for the initial land surface. *Loess_Tablelands_v2.py* generates a flat surface (Flat scenario) and optionally adds four closed depressions to it (Flat-Depressions scenario. Random roughness is added in either case. The initial flat surface used by v2 is set at the average elevation of the two bedrock tableland surfaces (1187.9 m); this was done to simplify visualization and has no real impact on model results.

Both scripts are organized to be run in three stages: 0-12,000, 12,000-16,000, and 16,000-26,000 model years. As described in the paper, these represent in simplified form the sequence of Peoria Loess deposition, Brady Soil formation during ongoing but much slower loess deposition, and Bignell Loess deposition at an intermediate rate (deposition here, as in the paper, means initial deposition prior to any erosion or deposition of reworked loess by hillslope and fluvial processes; accumulation is the net build-up of loess).

Parameters K_{sp} , D , and ω_{cr} , used in each stage for the LinearDiffuser and ErosionDeposition components, were selected based on a sensitivity analysis and our current understanding of environmental conditions during Brady Soil formation (see Section 2.5.3 of the paper and Supporting Information Text S5). For each stage, the script sets initial parameters and then loops through the appropriate number of 0.2 yr timesteps, also tallying erosion and deposition independent of loess deposition. Every 1250 timesteps (250 model yr), rasters of elevation and average erosion/deposition along with images of elevation and erosion are written.

Model parameters K_{sp} , D , ω_c , and “uplift rate” (=loess deposition rate) are ramped up or down gradually from first stage values to those of the second stage, starting 1200 years before the end of that stage, then ramped to the third stage values starting 1000 years before the end of the second stage. For v2, ω_c is 0 for the first 1200 yr of the first stage, then ramped up to its full value over the next 2400 yr. This was found to be necessary for drainage network initiation in the Flat and Flat-Depressions scenarios.

Text S4. Workflows used in analyzing topographic data and preparing figures for this paper

We used R (R Core Team, 2022) for most analysis and visualization of data; ArcGIS Pro (ESRI, Inc.) and QGIS (QGIS Development Team, 2024) were used in some cases as noted in the paper text or supporting text. All R scripts used are included as supporting data in Data Set 3 and also archived at Mason (2025), <https://doi.org/10.5281/zenodo.14867711>; developed in various versions of R, they all work in R 4.4.2 with the package versions listed here.

convert_to_tif.R and *convert_to_txt.R* can be used to convert between the ESRI ascii text DEMs that are output from the landscape evolution model and the more compact tiff format in which these DEMs are archived (terra 1.8-42)

In a DEM of a real landscape or model output, *drainage_net_analysis.R* identifies closed depressions and generates a drainage network, using GRASS GIS and Whitebox tools within QGIS, via the *qgisprocess* R package, plots results, and saves them as raster or vector files (terra 1.8-42, *qgisprocess* 0.4.1, *sf* 1.0-20, *rgrass* 0.5-2, *RColorBrewer* 1.1-3). Used for Figure 4.

Using erosion rate maps output every 250 model years by the landscape evolution model, *model_erosion_analysis.R* will calculate and plot cumulative erosion/deposition and fraction of deposited loess remaining as domain means or point values, over the 26,000 yr modeled or for one output time (terra 1.8-42, *qgisprocess* 0.4.1, *ggplot2* 3.5.2). Used for Figures 7, 9, S3.

slope_analysis_new_2.R uses DEMs output every 250 model years by the landscape evolution model, DEMs for individual output DEMs, or individual DEMs of real landscapes to produce slope maps. Also analyzes slopes by 10 m elevations slice but these results are not used in this paper (terra 1.8-42, *qgisprocess* 0.4.1, *sf* 1.0-20, *ggplot2* 3.5.2, *RColorBrewer* 1.1-3). Used for Figure 9.

geomorphons_d_density.R carries out a geomorphons analysis on all output DEMs from a landscape evolution model run, individual output DEMs, or DEMs of real tablelands. The geomorphons analysis is followed by calculation and mapping of the Valley Density Index and Domain Valley Density Index as described in the paper text (terra 1.8-42, *qgisprocess* 0.1.0, *sf* 1.0-20, *rgrass* 0.3-9, *RColorBrewer* 1.1-3). Used for Figures 7, 10, S4.

ldw2.R uses the *gstat* package to create interpolated surfaces from shapefiles of registered wells and test holes by the IDW method, and evaluates performance of a range of IDW parameters using LOOCV. Produces a table of RMSE for a range of IDW parameters, and then a final interpolated surface using the selected parameters. LOOCV is then used to calculate residuals and overall RMSE for the selected parameters, and

adds residuals to a new shapefile of wells and test holes (terra 1.8-42, sf 1.0-20, gstat 2.1-3, dismo 1.3-16, ggplot2 3.5.2). Used for Figure 3.

combine_plot_dvdi.R combines data from text files of DVDI produced by *geomorphons_d_density.R* and produces DVDI plots for all cases used in the sensitivity analysis (ggplot2 3.5.2). Used for Figure 7.

hypsometric3.R reads model output DEMs for selected time points and produces hypsometric curves (ggplot2 3.5.2, terra 1.8-42). Use in Figures 7, S5, S6.

rayshader_model_animation2.R creates a 3D perspective view of the modeled landscape at each 250 model yr output time, then combines these as frames in an animation.

rayshader_model_single_stage.R creates a 3D perspective view for a single output time (terra 1.7-46, hillshader 0.1.0, rayshader 0.35.7, rayrender 0.29.6, av 0.8.4). Used for Figures 6, 7, S1, S2.

Text S5. Sensitivity analysis

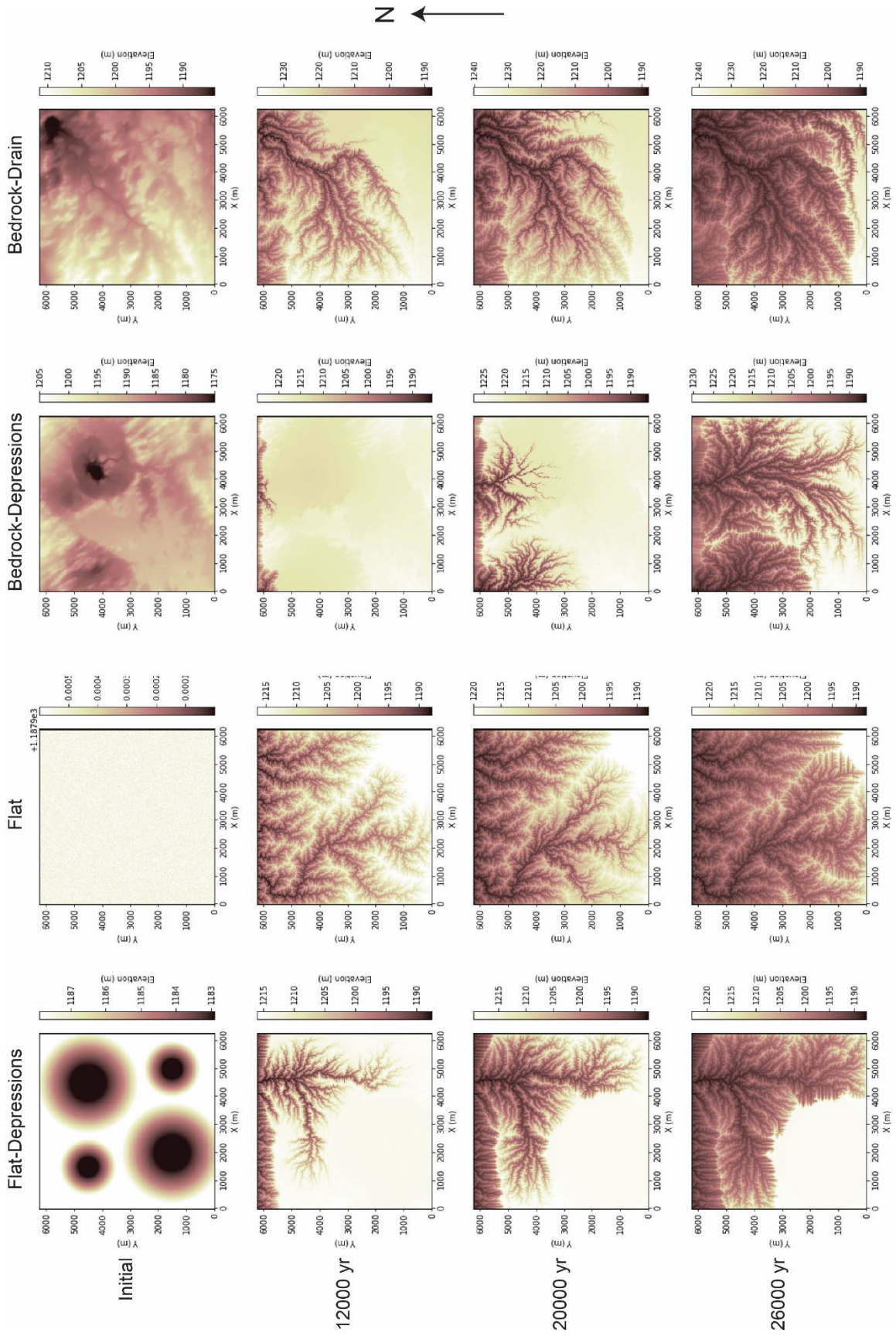
To systematically assess sensitivity of modeling results to the values of the key parameters, we carried out simulations as described in the paper text (26,000 yr, same four scenarios with differing initial topography, same loess accumulation rates by time interval), varying the values of K_{sp} , D , and ω_c . For all but one of these sensitivity tests, we changed those parameters by the same proportions as described in the paper (K_{sp} reduced by 50%, D and ω_c increased by 100%). The one exception was a simulation with $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$, $D = 0.05 \text{ m}^2\text{yr}^{-1}$, and $\omega_c = 0.001 \text{ m yr}^{-1}$ throughout the 26,000 yr (No Brady Soil Interval). The other sensitivity test simulations reported here include:

- 1) Four values of K_{sp} , 0.0004, 0.0008, 0.0012 (used in the paper), and $0.0016 \text{ m}^{-1/2}\text{yr}^{-1/2}$, with D held at $0.05 \text{ m}^2\text{yr}^{-1}$ and ω_c at 0.001 m yr^{-1} .
- 2) Two lower values of ω_c : a) $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$, $D = 0.05 \text{ m}^2\text{yr}^{-1}$, and $\omega_c = 0.0001 \text{ m yr}^{-1}$, b) $K_{sp} = 0.0008 \text{ m}^{-1/2}\text{yr}^{-1/2}$, $D = 0.05 \text{ m}^2\text{yr}^{-1}$, and $\omega_c = 0.0 \text{ m yr}^{-1}$. We assumed from prior experimentation that higher values of ω_c would reduce drainage density well below observed levels and that combining $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$ with $\omega_c = 0.0 \text{ m yr}^{-1}$ would produce erosion rates much too high to fit our goals.
- 3) The No Brady Soil Interval test, all parameters the same as in the paper except no reduction of K_{sp} or increase of D and ω_c during the Brady Soil Interval. Loess accumulation rates and other parameters are the same as in the paper.
- 4) One lower and two higher values of D : a) $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$, $D = 0.01 \text{ m}^2\text{yr}^{-1}$, $\omega_c = 0.001 \text{ m yr}^{-1}$, b) $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$, $D = 0.1 \text{ m}^2\text{yr}^{-1}$, $\omega_c = 0.001 \text{ m yr}^{-1}$, c) $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$, $D = 0.5 \text{ m}^2\text{yr}^{-1}$, $\omega_c = 0.001$.

Results of these sensitivity tests are shown in Figures S3, S4, S5, and S6 in this document, using plots of cumulative domain mean erosion and DVDI over time, and hypsometric curves at three timepoints, all for the four standard scenarios.

Interpretation of sensitivity tests: The sensitivity analysis supports our selection of $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$, $D = 0.05 \text{ m}^2\text{yr}^{-1}$, and $\omega_c = 0.001 \text{ m yr}^{-1}$ for the times of Peoria and Bignell Loess accumulation (Table 1), though there is some flexibility for each parameter. Extensive tablelands persist after 26,000 yr in all scenarios with $K_{sp} = 0.0004 \text{ m}^{-1/2}\text{yr}^{-1/2}$ and to a lesser extent with $K_{sp} = 0.0008 \text{ m}^{-1/2}\text{yr}^{-1/2}$, and peak DVDI is much lower than observed values for the former. $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$ and $0.0016 \text{ m}^{-1/2}\text{yr}^{-1/2}$ result in tableland summits occupying < 35% of the domain at 26,000 yr for the scenarios with depressions and almost none of the domain for the other two scenarios. DVDI and sediment yield are within or not far above the observed ranges for $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$ but well above them for $K_{sp} = 0.0016 \text{ m}^{-1/2}\text{yr}^{-1/2}$. With $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$, setting D at either 0.05 or 0.10 m^2yr^{-1} achieves our goals, but $D = 0.50 \text{ m}^2\text{yr}^{-1}$ resulted in much less tableland dissection and lower than observed peak DVDI. $D = 0.01 \text{ m}^2\text{yr}^{-1}$ resulted in much higher than observed DVDI. With $K_{sp} = 0.0012 \text{ m}^{-1/2}\text{yr}^{-1/2}$ and $\omega_c = 0.0001 \text{ m yr}^{-1}$ sediment yields and DVDI are too high. A combination of $K_{sp} = 0.0008 \text{ m}^{-1/2}\text{yr}^{-1/2}$ and $\omega_c = 0 \text{ m yr}^{-1}$ would achieve our goals but is inconsistent with the observation of distinct channel heads some distance from the upper end of small basins, and the edges of tableland summits are not as well-defined because of dissection extending farther upslope. As discussed in the paper text, we based parameters used during the Brady Soil interval on our current understanding of environmental conditions at that time, and do not have quantitative observations to compare with the modeled landscape during that interval.

Figure S1. (following page) Version of Figure 6 showing modeled landscapes in map view. Evolution of modeled loess tablelands from four different initial surfaces. Top row is initial surface, 12000 yr is at end of rapid Peoria Loess accumulation, 20000 yr is during slower Bignell Loess accumulation, and 26000 yr is end of simulation.



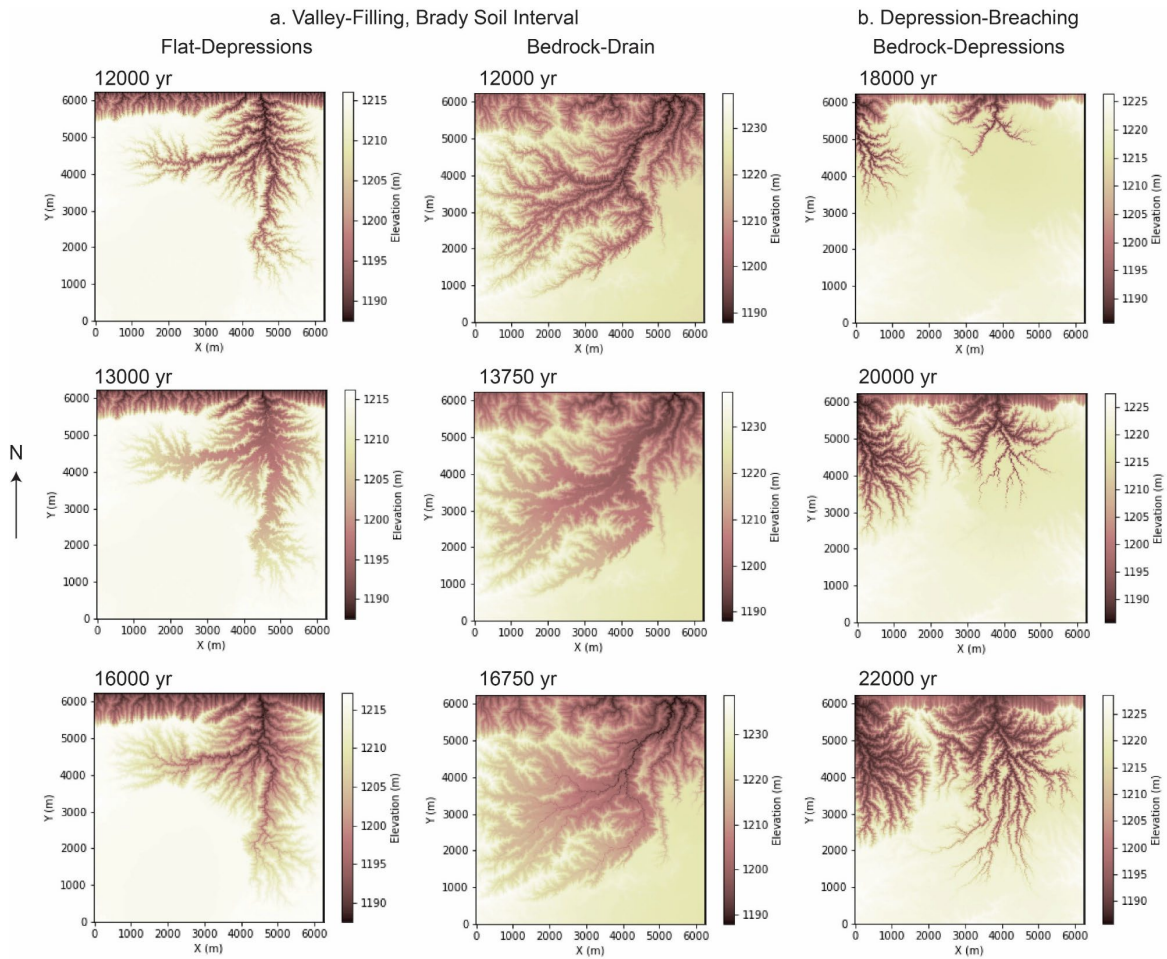


Figure S2. Version of Figure 7 showing modeled landscapes in map view. Key aspects of modeled loess tableland evolution. a. Valley filling during Brady Soil interval, followed by incision with change back to original model parameters, for two scenarios. Sequence is similar for Flat scenario; timing of adjustments are somewhat different for each scenario. b. Breaching of two closed depressions in the Bedrock-Depressions scenario. Similar breaching occurs in Flat-Depressions.

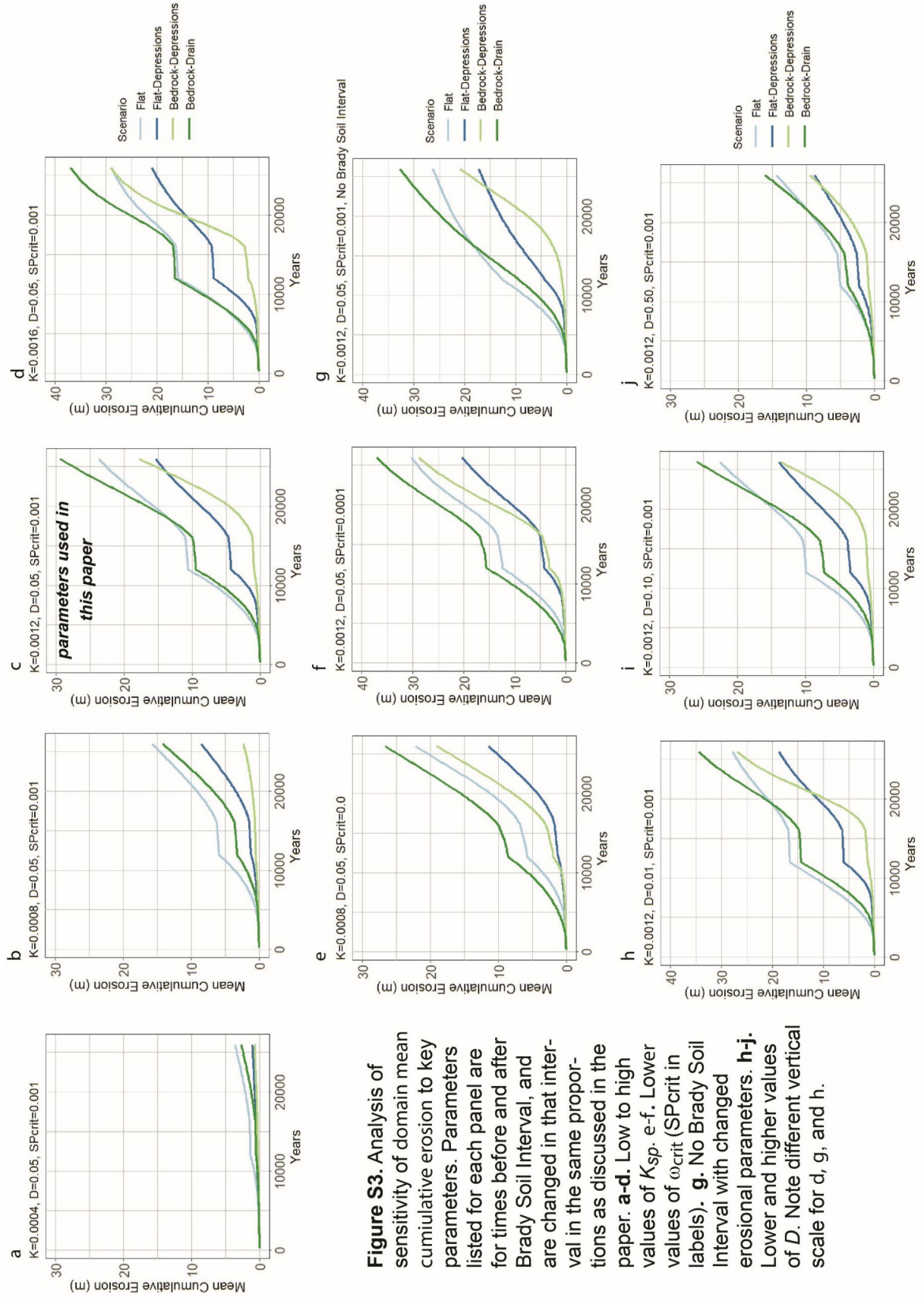


Figure S3. Analysis of sensitivity of domain mean cumulative erosion to key parameters. Parameters listed for each panel are for times before and after Brady Soil Interval, and are changed in that interval in the same proportions as discussed in the paper. **a-d.** Low to high values of K_{sp} . **e-f.** Lower values of ω_{crit} (SP_{crit} in labels). **g.** No Brady Soil Interval with changed erosional parameters. **h-j.** Lower and higher values of D . Note different vertical scale for d, g, and h.

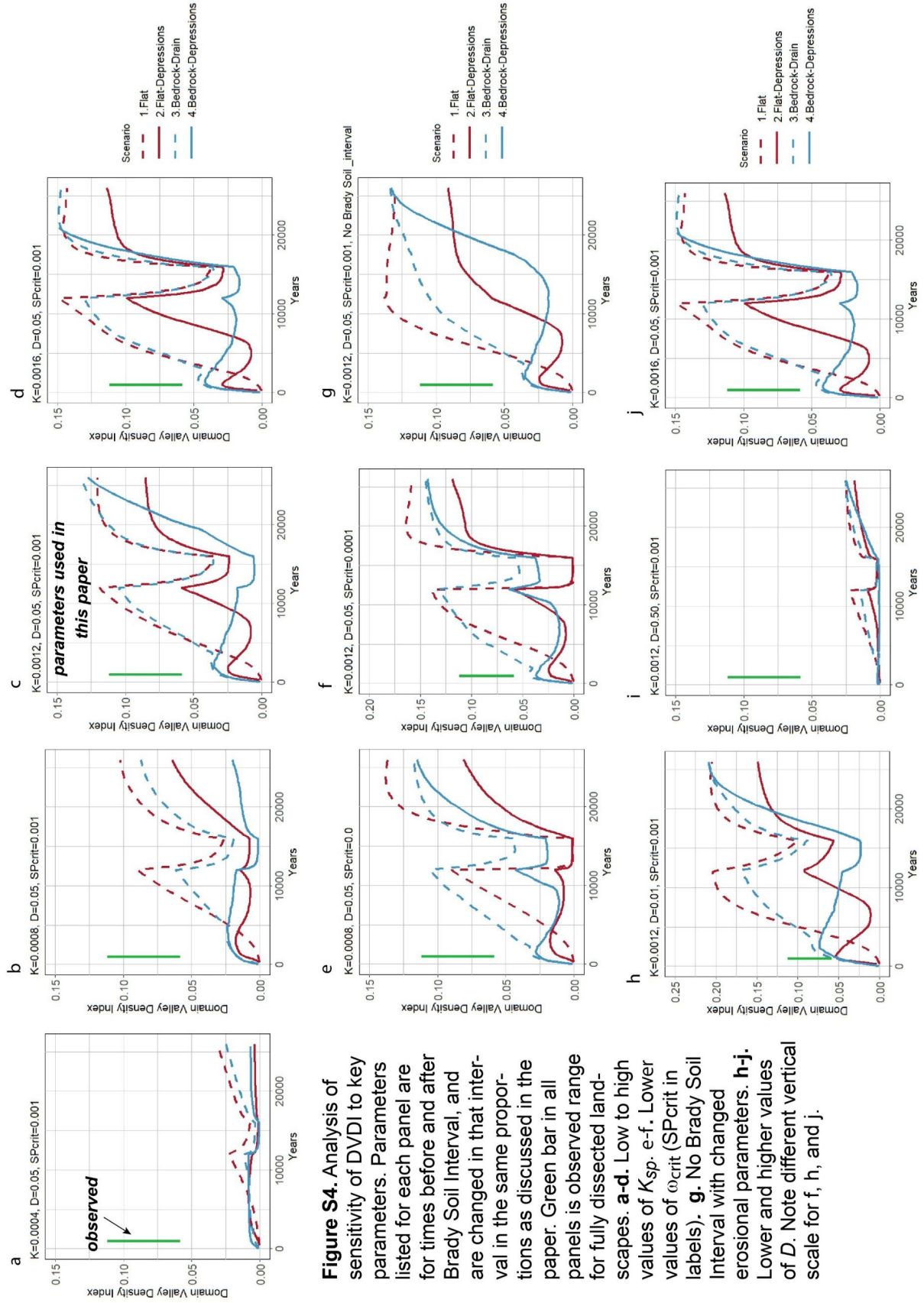
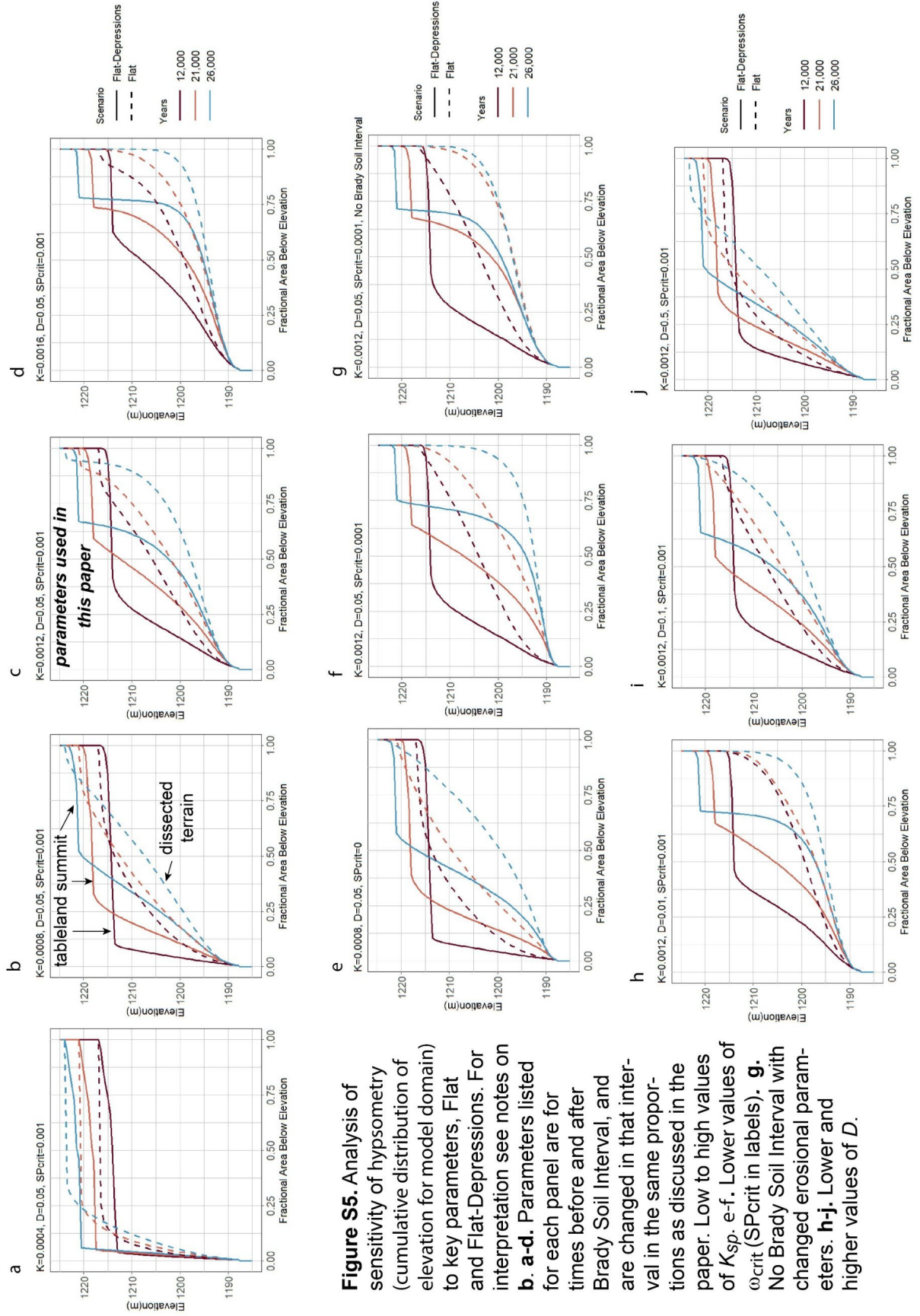


Figure S4. Analysis of sensitivity of DVDI to key parameters. Parameters listed for each panel are for times before and after Brady Soil Interval, and are changed in that interval in the same proportions as discussed in the paper. Green bar in all panels is observed land-scapes. **a-d.** Low to high values of K_{sp} . **e-f.** Lower values of ω_{crit} (SP_{crit} in labels). **g.** No Brady Soil Interval with changed erosional parameters. **h-j.** Lower and higher values of D . Note different vertical scale for **f**, **h**, and **j**.



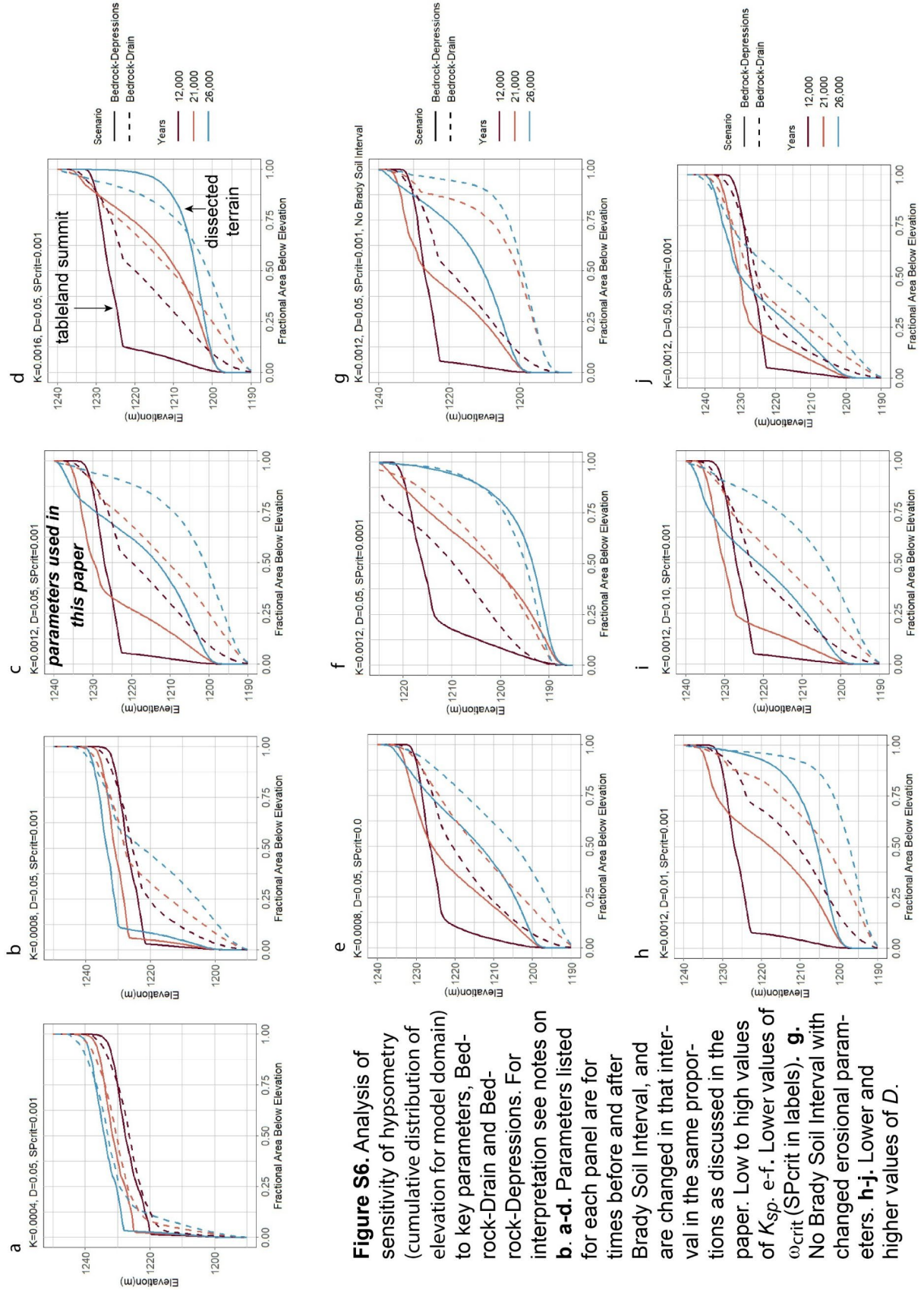


Figure S6. Analysis of sensitivity of hypsometry (cumulative distribution of elevation for model domain) to key parameters, Bedrock-Drain and Bedrock-Depressions. For interpretation see notes on **b. a-d.** Parameters listed for each panel are for times before and after Brady Soil Interval, and are changed in that interval in the same proportions as discussed in the paper. Low to high values of K_{sp} . **e-f.** Lower values of ω_{crit} ($SPcrit$ in labels). **g.** No Brady Soil Interval with changed erosional parameters. **h-j.** Lower and higher values of D .

Table S1. Dry bulk density and depth of Peoria Loess samples from central and western Nebraska (uploaded as separate file)

Table S2. Field strength measurements plotted in Figure 5 (uploaded as separate file)

Table S3. Maximum domain sediment yield (tonne km⁻² yr⁻¹, sensitivity analysis)

Parameters	Flat	Flat-Depressions	Bedrock-Drain	Bedrock-Depression
K=0.0004,D=0.05, Spc=0.001	402	102	397	63
K=0.0008,D=0.05, Spc=0.001	1724	1207	1781	471
K=0.0008,D=0.05, Spc=0.0	2570	1777	2557	2601
K=0.0012,D=0.01, Spc=0.001	2650	3328	3485	4751
K=0.0012,D=0.05, Spc=0.001	2769	1890	3000	3245
K=0.0012,D=0.05, Spc=0.0001	3008	2503	3463	4010
K=0.0012,D=0.10, Spc=0.001	2592	1555	2799	2952
K=0.0012,D=0.50, Spc=0.001	1415	1000	1971	1654
K=0.0016,D=0.05, Spc=0.001	3729	2951	3733	5167
K=0.0012,D=0.05, Spc=0.001, No Brady	3049	1584	2812	3130

Table S4. DVDI values for 10 fully dissected areas of 6260m X 6260 m in thick loess of the study area, with coordinates of northwest corner.

	Lat	Lon	DVDI
1	41.3475	-99.1537	0.079377
2	40.9138	-100.475	0.111839
3	40.9125	-100.549	0.101198
4	40.9757	-100.624	0.101693
5	41.5493	-99.6636	0.084254
6	41.6113	-99.6642	0.058363
7	41.2466	-99.4691	0.082506
8	41.2819	-99.6281	0.070974
9	41.2466	-99.4692	0.075384
10	41.2361	-99.9969	0.074233

Data Set S1. Registered wells and geologic test holes used to generate interpolated bedrock surfaces shown in Figure 3. Includes three shapefiles. See README.md in .zip file for details of attribute fields (**ds01**).

Data Set S2. Python scripts Loess_Tablelands_v1.py, used in landscape evolution modeling starting with flat surfaces and Loess_Tablelands_v2.py, used in landscape evolution modeling starting with real bedrock tableland surfaces, as described in Text S3,

together with initial surfaces for Bedrock-Depressions and Bedrock-Drain scenarios (**ds02**).

Data Set S3. Includes nine R scripts, used as described in Text S4 and S4 (**ds03**).

Movie S1. Animation of landscape evolution through 26,000 model years for Flat scenario. 3D perspective view is from northeast to southwest, viewed across north boundary which is open to water and sediment while others are closed. One frame/250 yr (**mov01**).

Movie S2. Animation of landscape evolution through 26,000 model years for Flat-Depressions scenario. 3D perspective view is from northeast to southwest, viewed across north boundary which is open to water and sediment while others are closed. One frame/250 yr (**mov02**).

Movie S3. Animation of landscape evolution through 26,000 model years for Bedrock-Drain scenario. 3D perspective view is from northeast to southwest, viewed across north boundary which is open to water and sediment while others are closed. One frame/250 yr (**mov03**).

Movie S4. Animation of landscape evolution through 26,000 model years for Bedrock-Depressions scenario. Perspective view is from northeast to southwest, viewed across north boundary which is open to water and sediment while others are closed. One frame/250 yr (**mov04**).